Changes in nitrogen emissions in the Netherlands

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Set-up your environment

```
## Loading required package: tidyverse

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
## Loading required package: rmarkdown
##
## Loading required package: yaml
```

Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

- Why is this relevant?
- Nitrogen emissions represent a critical public health crisis in the Netherlands. Elevated atmospheric concentrations lead to the production of particulate manner and ozone, which are closely associated with serious health problems such as worsening asthma, long-term cardiovascular problems, and diminished lung function.
- This report measures the correlation between public health concerns and regional nitrogen levels over almost 20 years, from 2005 to 2023. By analyzing these temporal and spatial patterns, we want to pinpoint pollution hotspots and offer a data-driven framework for focused reduction initiatives, like industrial restrictions or localized agriculture changes, to successfully lessen the burden on public health.

Part 2 - Data Sourcing

2.1 Load in the data

Here we imported the raw data regarding air pollution, particulate matter, nitrogen sources, and monitoring locations into our environment from their original CSV formats. Once the data is loaded, these files are saved as compressed RDS files. Moving from plain text files to R's own format keeps the data organized and locks all variable types in place. This makes sure the files load much faster and work correctly in future steps.

2.2 Provide a short summary of the dataset(s)

The data set PolutionInGeneral contains 11 variables, each representing a different year. For each year, data is provided on the levels of nitrogen, NMVOCs (non-methane volatile organic compounds), sulfur oxides, and PM2.5 particulate matter observed in the air. By analyzing this data, we can assess the severity of air pollution and identify trends in air quality over time.

The data set JaargemiddeldeFijnstof show particulate matter (fine dust) levels measured at various monitoring stations across the Netherlands over the years 2005 till 2025. This provides a clear insight into the periods and regions where particulate matter causes the greatest impact and pollution.

The data set shows the sources of nitrogen measured over a time period from 2005 till 2024

The data set AA_BronnenVanFijnstof shows the sources of fine dust measured over a time period from 2005 till 2024

The data set Meetlocaties is used to find the locations of the Annual Average Fine Dust so we can pin it to the correct places on the dutch map.

2.3 Describe the type of variables included

Most of the datasets include a duration of time most often 2005 till 2024, and a variable that may be a specific measuring station or a source of polution.

Part 3 - Quantifying

3.1 Data cleaning

In this step we cleaned both the particulate matter and nitrogen datasets. First, the very last row is removed from each file to get rid of unnecessary summary text. Next, the numbers are fixed by changing European commas into standard periods so that R can read them correctly as numeric data. Finally, a new total column is created for each dataset by adding up the emissions from all the individual sectors.

Cleans the Meetlocaties dataset by splitting all data into separate columns and removing the first four rows.

We also cleaned the Meetlocaties dataset by splitting all data into separate columns and removing the first four rows.

Puts Meetlocaties & JaargemiddeldeFijnstof together the data gets a clear location

After that we merges particulate matter measurements with monitoring location data, adds geographic information, and cleans up the resulting dataset by keeping only the most relevant columns.

Makes Fijnstof_Met_Locaties numeric

This code converts coordinates to numeric values and cleans all yearly measurement columns by removing special characters and converting them to numbers.

3.2 Generate necessary variables

Creates total variables

In this step, the total annual emissions for both particulate matter and nitrogen are pulled into two smaller, simple tables. The code then matches and merges these two tables together into a single file based on the year. Because the nitrogen data ends in 2023, the combined table automatically stops at the year 2023 as well.

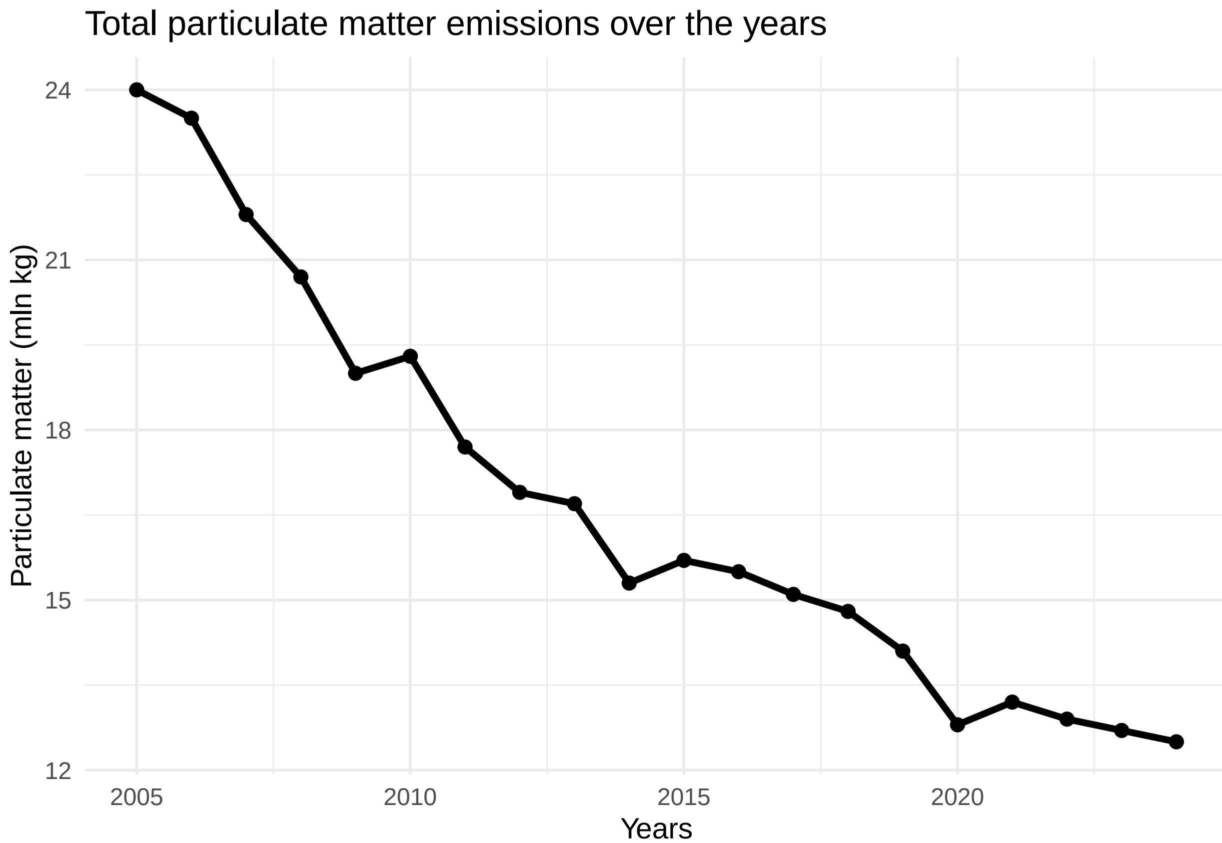
DIstancing data from the randstad from the Non-randstad

Here we have combined a list of specific cities to form define the Randstad region, The code then checks the city name for each monitoring station: if it is on the list, it gets labeled as Randstad, and if not, it is labeled as Other regions.

3.3 Create line graphs

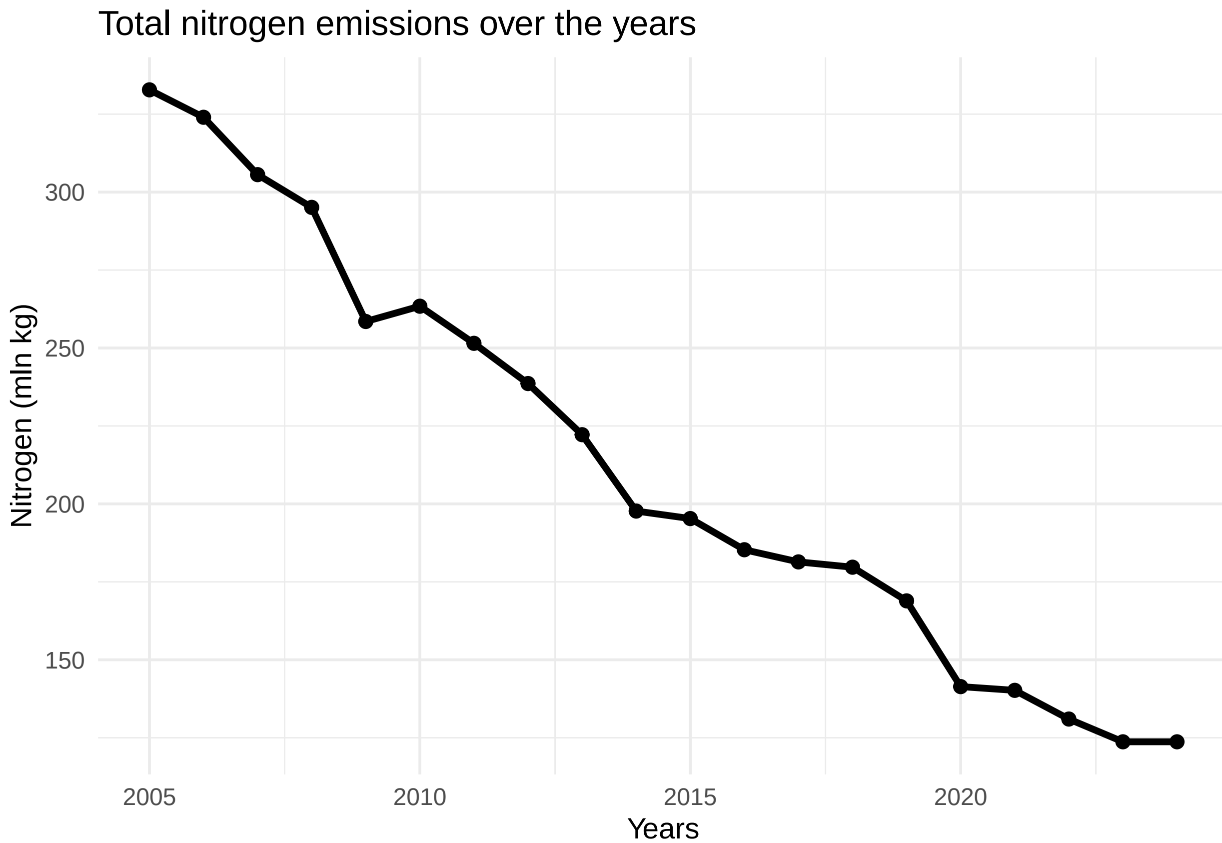
Simple graph about the general emission of particulate matter (fijnstof)

This graph shows the evolution of total particulate matter emissions over the years from 2005 to 2025, where you can see that the total particulate matter emissions have almost halved over the years.



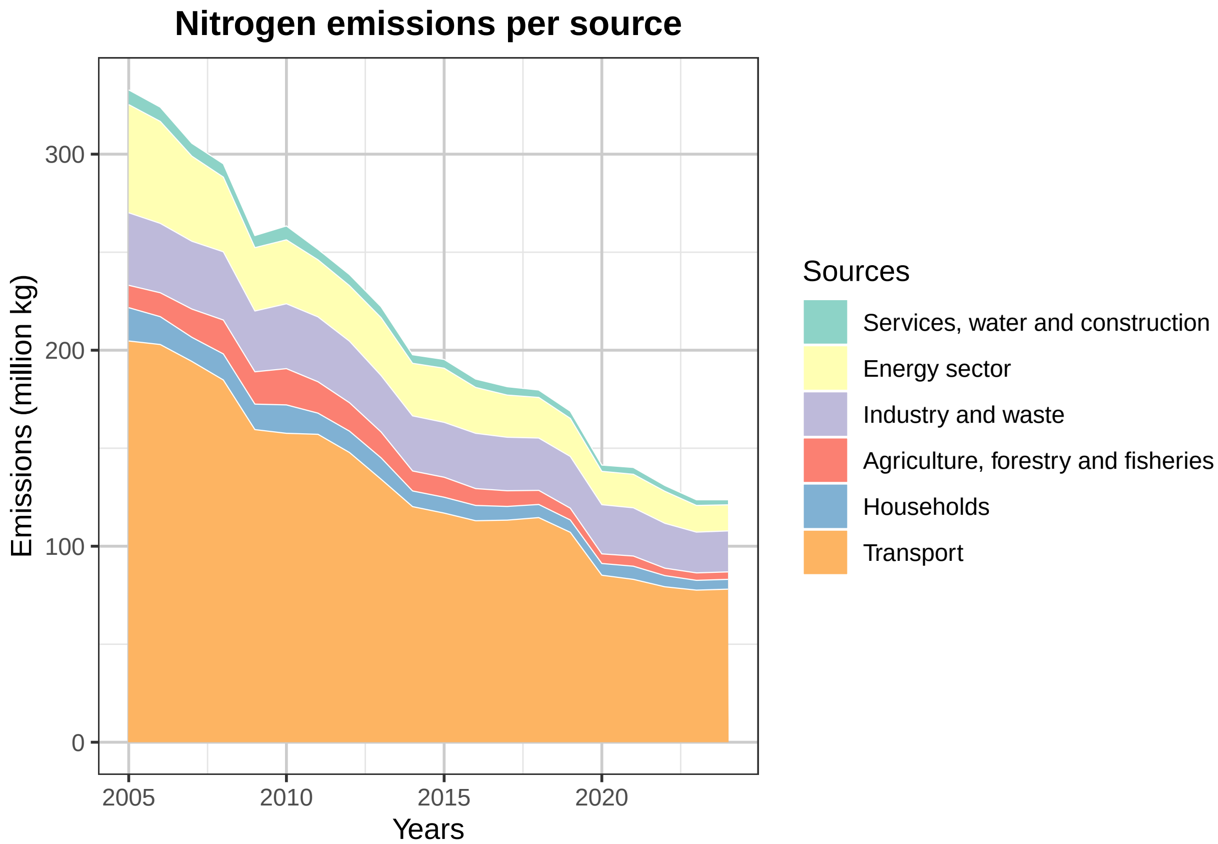
Simple graph about the general emission of nitrogen

This graph shows the evolution of the emission of nitrogen over the years from 2005 to 2025, where you can see that the total particulate matter emissions have more than halved over the years.



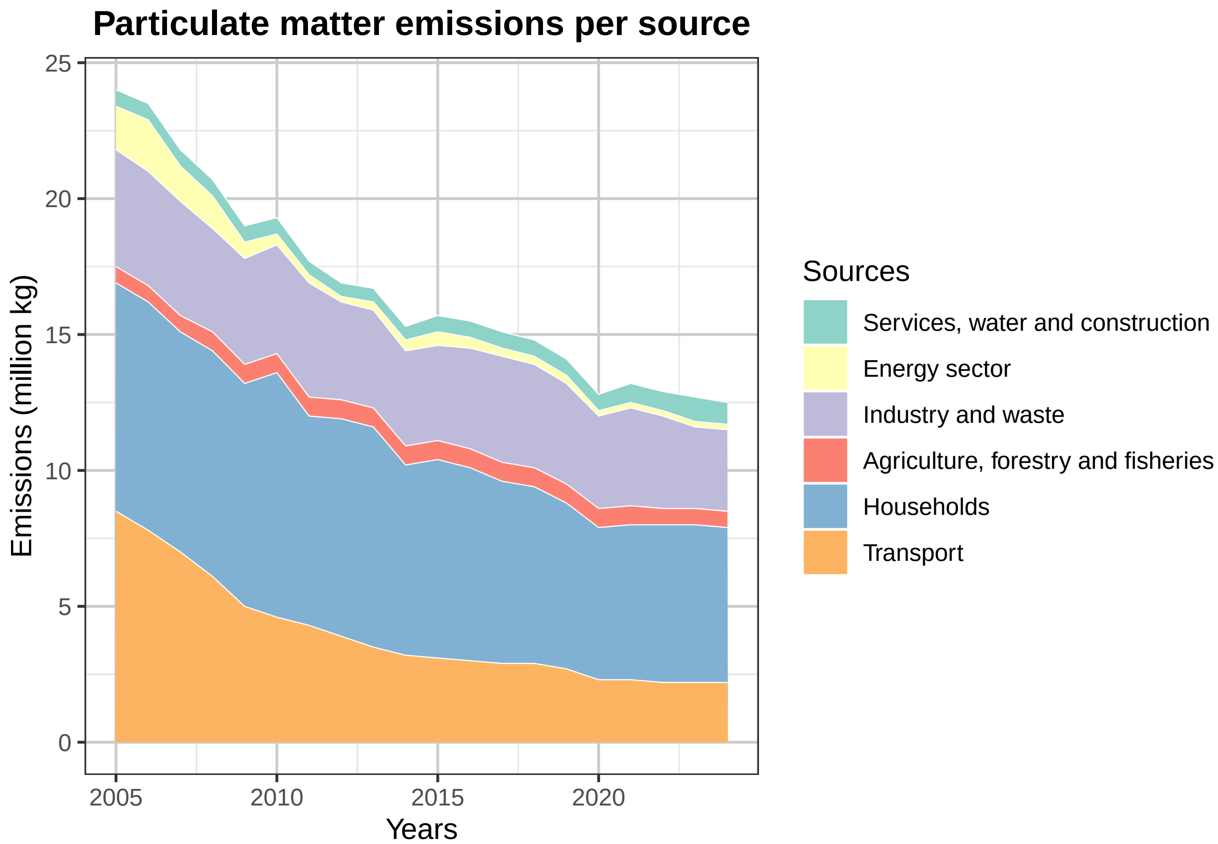
Detailed and Multifaceted Map of Nitrogen Emission Origins

A more detailed graph displays the different sources of nitrogen emissions using distinct colors, making it easier to distinguish between the various emission sources.



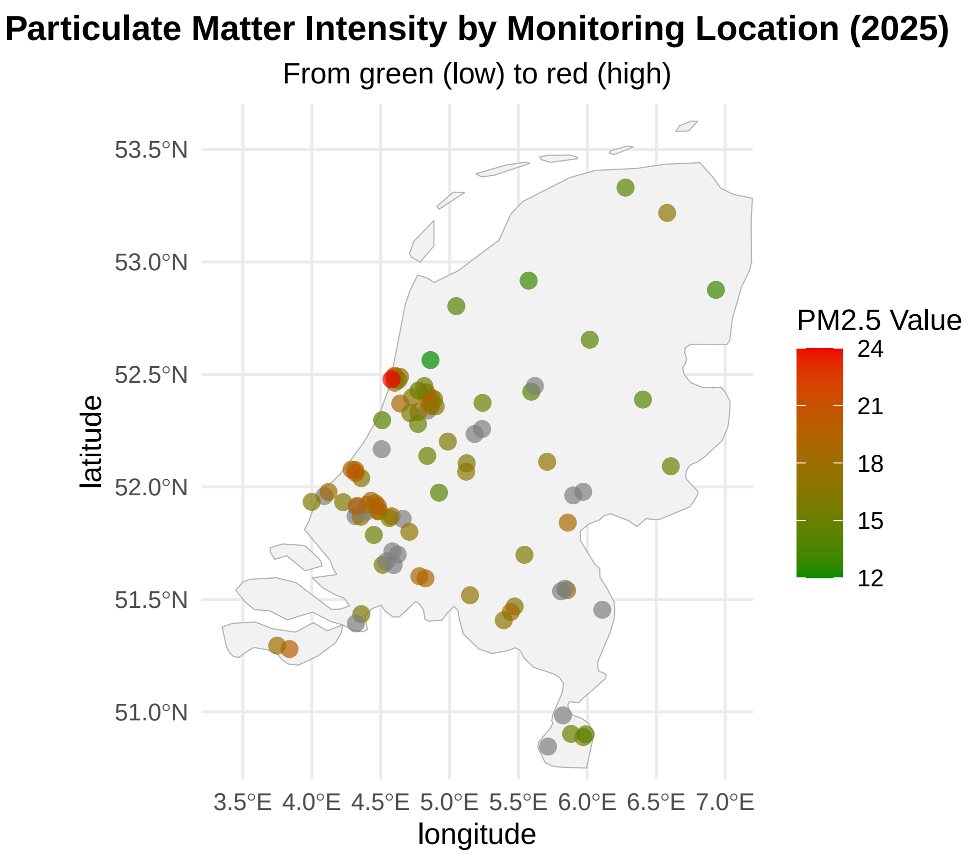
Detailed and Multifaceted Map of Particulate matter Origins

A more detailed graph displays the different sources of fine particulate matter emissions using distinct colors, making it easier to distinguish between the various emission sources.



3.4 Creating a heatmap

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

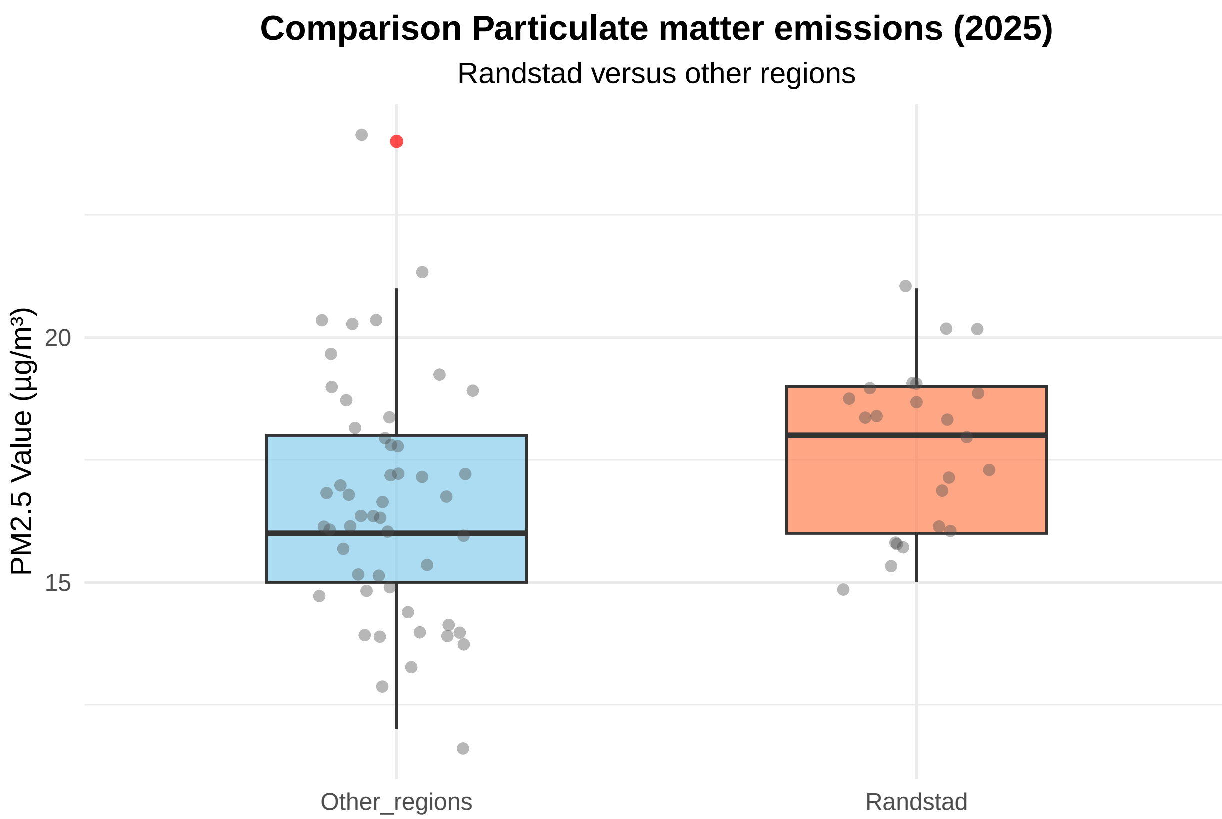


This map displays the locations of air quality monitoring stations across the Netherlands and shows the measured concentrations of fine particulate matter in the air for the year 2025. A conventional heatmap, in which all municipalities are represented by different colors, was not feasible because data were available for only a limited number of municipalities. In addition, larger municipalities often contained multiple monitoring stations, which sometimes recorded substantially different measurements due to local environmental conditions and station-specific locations. As a result, displaying the individual monitoring stations provides a more accurate representation of the available data.

3.5 Visualize sub-population variation

A boxplot that combines the randstad and the rest

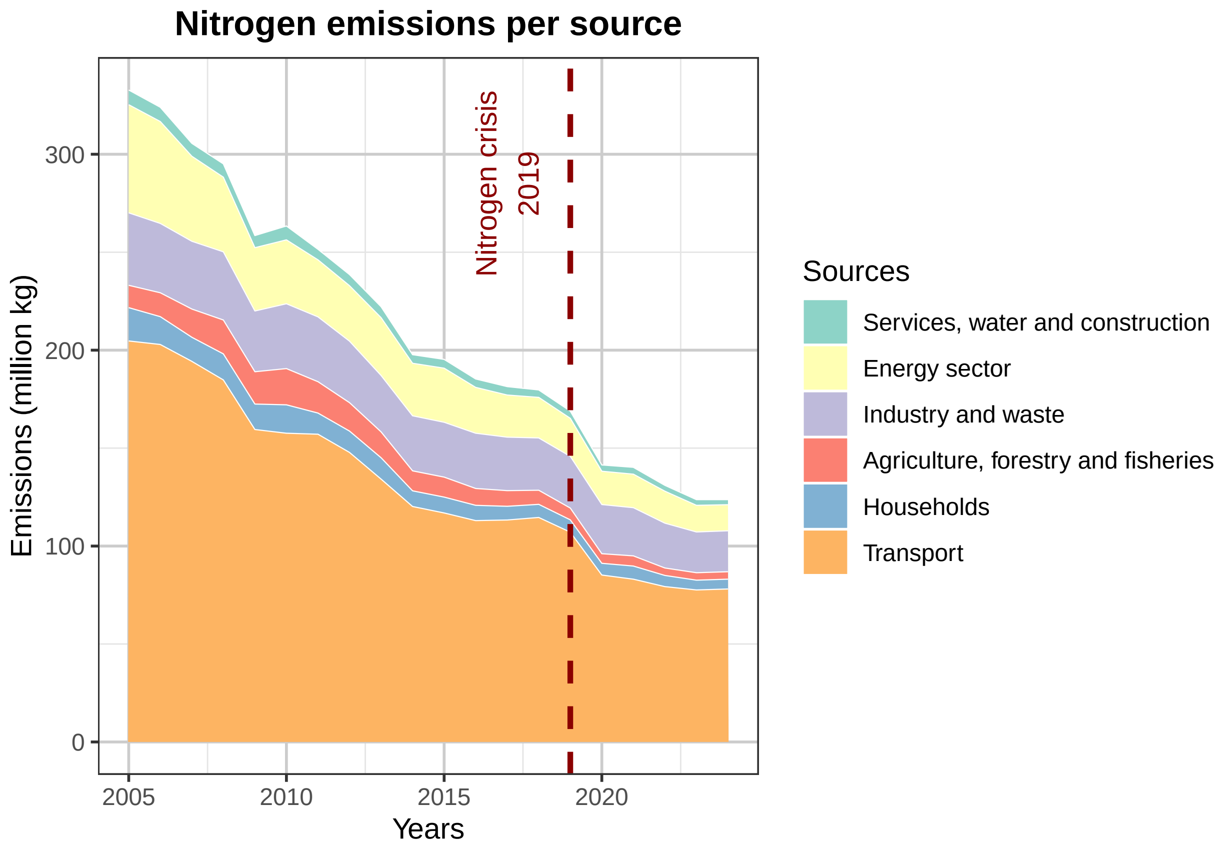
```
## Scale for x is already present.
## Adding another scale for x, which will replace the existing scale.
```



By comparing the PM 2.5 particulate matter levels between the randstad boxplot and the other regions boxplot, the boxplot comparison clearly shows regional differences in Dutch air quality. This graphic directly relates to the project's main social issue, which is examining the effects of transportation, industrial infrastructure, and densely populated areas on localized pollution and public health.

These plots demonstrate that randstad areas have a greater, more concentrated load of particle matter, even though national data indicates that industries like agriculture dominate nitrogen emissions across rural regions. The randstad boxplot's larger median PM 2.5 value highlights the unequal distribution of environmental effects and presents air pollution as a pressing public health and urban development issue.

3.6 Event analysis



The graph shows the development of nitrogen emissions from different sources in the Netherlands between 2005 and 2024. Overall, emissions decreased substantially over the observed period, with the largest contribution consistently coming from the transport sector. Emissions from industry, the energy sector, households, agriculture, and services also show a downward trend.

The vertical dashed line indicates the Dutch nitrogen crisis in 2019. After this point, emissions continued to decline across most sectors. This trend coincides with the introduction of stricter environmental measures and, shortly afterwards, the COVID-19 pandemic. However, based on this graph alone, no direct causal relationship can be established between these events and the observed decrease in emissions.

In the graph you see that there is a strong drop in nitrogen emissions across multiple sources from 2019 onward. This can be explained by multiple factors, such as new more strict emissions laws by the Dutch government and also the EU. These laws include a lowering of the speed limit to 100 km/h, causing a sharp drop in vehicle emissions. The laws also include the freezing of a lot of infrastructure and other building projects causing a drop in services and industry emissions. And then from late 2019 and 2020 onward there was the COVID pandemic causing lower emissions in almost all sectors.

Part 4 - Discussion

4.1 Discuss your findings

The comparative boxplots show that the median level of fine particulate matter emissions is higher in the Randstad region than in non-Randstad areas. This pattern is also reflected in the map visualization, where a high concentration of red dots can be observed within the Randstad.

Furthermore, the data indicate that fine particulate matter emissions have decreased since 2005. This downward trend is visible across all of our graphs and can also be observed in the map visualization when the displayed year is adjusted in the code.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/OLHHRuyten/Air-pollution-and-health-4.3-NL

5.2 Reference list

AA_BronnenVanFijnstof, BA_BronnenVanStikstof and PolutionIngeneral are from CBS:

Centraal Bureau voor de Statistiek. (2025, 2 september). Uitstoot luchtverontreinigende stoffen lager dan norm voor 2030. https://www.cbs.nl/nl-nl/nieuws/2025/36/uitstoot-luchtverontreinigende-stoffen-lager-dan-norm-voor-2030 Dataset JaargemiddeldeFijnstof Planbureau voor de Leefomgeving. (z.d.). Atlas van de Regio. https://themasites.pbl.nl/atlas-regio/